

Project Topic and Scope

Multimodal Behavioral Cloning for a Robotic Arm Using Mamba and OpenCLIP

Oria Cohen | Computer Science Final Project

Project Type	Offline multimodal robotic imitation learning pipeline
Main Task	Next-action prediction using behavioral cloning
Core Technologies	YOLO, OpenCLIP, Mamba, PyTorch, Streamlit
Dataset Focus	Recorded BridgeData-style robotic manipulation trajectories, including a selected large subset of approximately 30GB
Final Output	Continuous 7-dimensional robot action vector

1. Project Topic

This project focuses on visual sequence learning for robotic imitation learning. The goal is to build an offline software pipeline that learns from recorded robotic demonstrations by combining image-sequence information with robot-state data and predicting the next robot action.

Instead of making decisions from a single image, the system uses a temporal window of consecutive observations. This allows the model to learn motion context, such as how the robot approaches an object, when the gripper interacts with it, and what action should follow next.

2. Problem Definition

Robotic manipulation data is complex because it combines visual input, robot-state signals, temporal dependencies, and noisy real-world demonstrations. A single frame may not be enough to understand the task state, while robot-state signals alone may miss visual failures such as an object slipping, an incorrect grasp, or an invalid trajectory.

The main challenge is therefore to create a reliable representation that combines what the robot sees with what the robot state reports, and to process this representation over time in an efficient way.

3. Final Project Scope

The final system is implemented as an offline research and engineering pipeline for multimodal behavioral cloning. It does not control a physical robot in real time. Instead, it operates on recorded trajectories, prepares the data, trains or loads a Mamba-based model, evaluates predictions, and displays the results in a Streamlit interface.

- Input: recorded robotic trajectory folders containing image frames and robot-state/action data.
- Main learning task: continuous next-action prediction.
- Visual representation: OpenCLIP embeddings computed from image frames.
- Robot representation: 7-dimensional robot-state vectors aligned with the visual embeddings.
- Sequence model: Mamba, used to learn temporal dependencies over multimodal input windows.
- Evaluation: quantitative error metrics and qualitative prediction examples.

4. Implemented Solution

The implemented solution is a staged pipeline that starts with dataset validation and ends with evaluation results visualization. A YOLO-based data curation stage was added in order to improve the quality of trajectories used for training. The YOLO work included manual image organization, Roboflow annotation, training/fine-tuning of a detector for the project environment, threshold tuning, and manual validation of filtering outputs.

After filtering, successful trajectories are stored in a filtered CSV file. Image frames from those trajectories are encoded with OpenCLIP into 512-dimensional visual embeddings. These embeddings are concatenated with 7-dimensional robot-state vectors, forming a 519-dimensional multimodal representation at each timestep. Fixed-length sequence windows are then used as input to a Mamba model that predicts the next 7-dimensional robot action.

5. Dataset and Filtering Focus

The project was developed around a large selected subset of BridgeData-style robotic manipulation trajectories. The original trajectory set contained approximately 8,000 trajectory folders. After applying the YOLO-based and robot-state filtering logic, approximately 4,100 trajectories were retained as successful examples for the downstream pipeline.

The filtering policy was intentionally conservative. In ambiguous cases, it was preferable to reject a trajectory that may have been successful rather than incorrectly accept a visually invalid or failed trajectory as a successful training example. This decision supports more reliable behavioral cloning training because the model learns directly from the filtered demonstrations.

6. Main Pipeline Stages

Stage	Purpose
Dataset Selection	Select and validate the trajectory dataset folder.
YOLO-Based Curation	Detect gripper/object conditions and filter visually invalid or unreliable trajectories.
Filtered CSV	Store the accepted trajectory list and status information for downstream stages.
OpenCLIP Embeddings	Encode each image frame into a compact 512-dimensional visual vector.
Multimodal Fusion	Concatenate visual embeddings with 7-dimensional robot-state vectors.
Mamba Sequence Modeling	Learn temporal dependencies and predict the next robot action.
Evaluation and Results	Compute metrics and display prediction examples, charts, tables, and logs.

7. Final Output

The final model output is a continuous 7-dimensional action vector. The system evaluates how close the predicted action is to the ground-truth robot action using metrics such as MAE, MSE, and per-dimension MAE, together with visual examples and prediction-vs-ground-truth charts.

8. Out of Scope

The final implementation focuses on offline next-action prediction from recorded data. Real-time closed-loop robot control, deployment on a physical robot, full benchmarking across all BridgeData tasks, and a complete experimental comparison against alternative sequence architectures are outside the final implementation scope.

9. Summary

The final project implements a complete multimodal robotic imitation learning pipeline. It combines YOLO-based data curation, OpenCLIP visual representation, robot-state fusion, Mamba sequence modeling, training, checkpoint-based evaluation, and Streamlit result visualization into one coherent system for offline behavioral cloning.